

MATH 231 – Linear Algebra I

Lecture 0 – Foundations Review

A self-study primer to read *before* the course begins

Fall 2026

Before you start

This lecture is **Lecture 0**: it contains no new linear algebra. Everything here is material you have seen before, gathered in one place so that on Day 1 we can speak a common language. Your only formal prerequisite is **MATH 141** (Calculus I). If a page feels like review, skim it; if a page feels new, slow down and work the examples with a pencil — that is the whole point of a Lecture 0.

How to use this document

Read actively. For every *Example*, cover the solution and try it first. The **Looking ahead** notes tell you *why* a piece of review matters — each points to the unit of MATH 231 where the idea returns with full force. A short self-check with answers is at the end.

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1. Numbers and Sets

1.1 The number systems

We build up from the counting numbers to the real line, each system fixing a shortcoming of the previous one.

$$\underbrace{\mathbb{N}}_{\text{naturals}} \subset \underbrace{\mathbb{Z}}_{\text{integers}} \subset \underbrace{\mathbb{Q}}_{\text{rationals}} \subset \underbrace{\mathbb{R}}_{\text{reals}} \subset \underbrace{\mathbb{C}}_{\text{complex}}$$

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ (in this course we include 0; some books start at 1).
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ — adds subtraction.
- $\mathbb{Q} = \{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\}$ — adds division.
- $\mathbb{R}$ — fills the “gaps” (e.g. $\sqrt{2}$, π , e are real but not rational).
- $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\}$ — adds roots of *every* polynomial.

Looking ahead. A vector space (Unit 1) is always built over a number system called its field of scalars — almost always $\mathbb{R}$ or $\mathbb{C}$. Complex numbers get their own treatment in Unit 3, where $i^2 = -1$ becomes rotation by 90° .

1.2 Sets, elements, and membership

A **set** is an unordered collection of distinct objects called its **elements**. We write:

$$x \in A \quad (“x \text{ is an element of } A”), \quad x \notin A \quad (“x \text{ is not an element of } A”).$$

Two ways to describe a set:

- **Roster (list) form:** $A = \{2, 4, 6, 8\}$.
- **Set-builder form:** $A = \{x : \text{condition on } x\}$, read “the set of all x *such that* ...”. The colon “:” (or “|”) means *such that*.

Example 1.1. Write “the even numbers between 1 and 9” in set-builder form.

$$A = \{x \in \mathbb{Z} : 1 \leq x \leq 9 \text{ and } x \text{ is even}\} = \{2, 4, 6, 8\}.$$

The part before the colon says *what kind* of object ($x \in \mathbb{Z}$); the part after says *which ones* we keep.

Example 1.2. The unit circle is a set of points: $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. This is the same object as the geometric circle of §7 — a *set* and an *equation* describing the same thing.

The **empty set** $\emptyset = \{ \}$ has no elements. The number of elements of a finite set A is its **cardinality** $|A|$; e.g. $|\{2, 4, 6, 8\}| = 4$.

Looking ahead. *Set-builder notation is how we will define the central objects of the course: a span, a subspace, a null space, and the feasible region of an optimization problem are all written $\{x : \text{some linear condition}\}$.*

1.3 Subsets and equality

A is a **subset** of B , written $A \subseteq B$, if every element of A is also in B . If in addition $A \neq B$ we call it a **proper subset**, $A \subsetneq B$. Two sets are **equal** exactly when each contains the other:

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A).$$

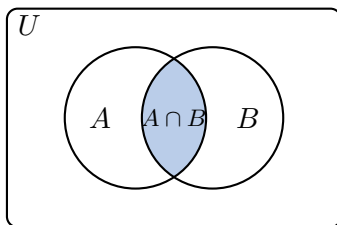
This “two-way inclusion” is the standard way to *prove* two sets are equal.

Example 1.3. Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$. Then $A \subseteq B$ (indeed $A \subsetneq B$), but $B \not\subseteq A$ since $3 \in B$ but $3 \notin A$. Also $\emptyset \subseteq A$ for *every* set A .

1.4 Set operations

Given sets A, B inside some universe U :

Union	$A \cup B = \{x : x \in A \text{ or } x \in B\}$	everything in either
Intersection	$A \cap B = \{x : x \in A \text{ and } x \in B\}$	the overlap
Difference	$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$	in A but not B
Complement	$A^c = U \setminus A = \{x \in U : x \notin A\}$	everything outside A



Example 1.4. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, universe $U = \{1, 2, \dots, 8\}$.

$$A \cup B = \{1, 2, 3, 4, 5, 6\}, \quad A \cap B = \{3, 4\}, \quad A \setminus B = \{1, 2\}, \quad A^c = \{5, 6, 7, 8\}.$$

The **Cartesian product** pairs elements from two sets in order:

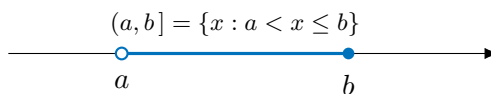
$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

The plane is $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}$ and space is $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$. Order matters: $(1, 2) \neq (2, 1)$.

Looking ahead. $\mathbb{R}^n = \mathbb{R} \times \dots \times \mathbb{R}$ (n copies) is where vectors live (Unit 1). A vector is nothing more exotic than an ordered list of n real numbers — an element of a Cartesian product.

1.5 Intervals and the real line

Intervals are the most common subsets of $\mathbb{R}$. They come in open, closed, and half-open flavors; a filled dot ($\bullet$) means the endpoint is included, an open dot ($\circ$) means it is excluded.



$[a, b] = \{x : a \leq x \leq b\}$	closed (both ends in)
$(a, b) = \{x : a < x < b\}$	open (both ends out)
$[a, b) = \{x : a \leq x < b\}, \quad (a, b] = \{x : a < x \leq b\}$	half-open
$[a, \infty) = \{x : x \geq a\}, \quad (-\infty, b) = \{x : x < b\}$	unbounded
$(-\infty, \infty) = \mathbb{R}$	the whole line

∞ is *not* a number, only a direction — so it is *always* paired with a parenthesis, never a bracket.

Example 1.5. Solve $|x - 3| < 2$ and write the solution as an interval.

$$|x - 3| < 2 \iff -2 < x - 3 < 2 \iff 1 < x < 5 \iff x \in (1, 5).$$

1.6 Absolute value and distance on the line

The **absolute value** $|x|$ is the distance from x to 0:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0. \end{cases}$$

The distance between two real numbers is $|a - b|$. Useful facts: $|xy| = |x||y|$, and the **triangle inequality** $|x + y| \leq |x| + |y|$.

Looking ahead. $|a - b|$ is distance in one dimension. In Unit 2 we generalize it to a norm $\|x - y\|$, which measures distance between vectors — and the triangle inequality survives unchanged.

2. Functions

A **function** $f : A \rightarrow B$ assigns to each input $x \in A$ exactly one output $f(x) \in B$. Here A is the **domain**, B the **codomain**, and

$$\text{range}(f) = \{f(x) : x \in A\} \subseteq B$$

is the set of values actually hit (the **image**).

Example 2.1. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$: the domain is $\mathbb{R}$, the codomain is $\mathbb{R}$, but the range is $[0, \infty)$ because squares are never negative.

Composition feeds one function into another: $(g \circ f)(x) = g(f(x))$ — do f first, then g . An **inverse** f^{-1} undoes f : $f^{-1}(f(x)) = x$. A function has an inverse exactly when it is one-to-one (distinct inputs give distinct outputs).

Example 2.2. If $f(x) = 2x + 1$ then $f^{-1}(x) = \frac{x-1}{2}$, since solving $y = 2x + 1$ for x gives $x = \frac{y-1}{2}$.
Check: $f^{-1}(f(3)) = f^{-1}(7) = 3$. ✓

Looking ahead. A linear transformation (Unit 8) is a special kind of function between vector spaces, and matrix multiplication is exactly composition of such functions. The matrix inverse (Unit 4) is a function inverse. Exponential and logarithm (§5) are inverses of each other.

3. Summation (Σ) and Product (Π) Notation

3.1 Sigma notation

Σ (capital sigma) is shorthand for a sum. The **index** i runs from the lower bound to the upper bound, and we add up the terms:

$$\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n.$$

The index name is a dummy: $\sum_{i=1}^n a_i = \sum_{k=1}^n a_k$.

Example 3.1. $\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30.$ $\sum_{k=0}^3 2^k = 1 + 2 + 4 + 8 = 15.$

Algebra of sums (use these constantly)

$$\begin{aligned} \sum_{i=1}^n (a_i + b_i) &= \sum_{i=1}^n a_i + \sum_{i=1}^n b_i && \text{(split a sum),} && \sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i && \text{(pull out constants),} \\ \sum_{i=1}^n c &= c n && \text{(a constant term added } n \text{ times),} && \text{Index shift: } \sum_{i=1}^n a_i &= \sum_{j=0}^{n-1} a_{j+1}. \end{aligned}$$

The first two properties together are exactly *linearity*, the single most important structural fact in the whole course.

Example 3.2 (Telescoping). A sum where neighbors cancel:

$$\sum_{i=1}^n (a_{i+1} - a_i) = a_{n+1} - a_1.$$

For instance $\sum_{i=1}^n \frac{1}{i(i+1)} = \sum_{i=1}^n \left(\frac{1}{i} - \frac{1}{i+1} \right) = 1 - \frac{1}{n+1}$.

Looking ahead. The dot product $\langle x, y \rangle = \sum_i x_i y_i$ (Unit 2), the matrix product, and every “cost of an algorithm” count (Unit 6) are written with Σ . Linearity of the sum is the reason these objects are called “linear.”

3.2 Pi notation and factorials

Π (capital pi) is the same idea for *products*:

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdots a_n.$$

The **factorial** is the product of the first n positive integers:

$$n! = \prod_{i=1}^n i = 1 \cdot 2 \cdot 3 \cdots n, \quad 0! = 1 \text{ (by convention).}$$

Example 3.3. $\prod_{i=1}^4 i = 4! = 24.$ $\prod_{k=1}^3 (2k) = 2 \cdot 4 \cdot 6 = 48.$

Looking ahead. The determinant of a matrix is a signed sum of products of entries (Unit 4), and equals the product of the eigenvalues $\det A = \prod_i \lambda_i$ (Unit 11). Factorials reappear in the binomial theorem (§6) and Taylor's theorem (Unit 5).

4. Common Summation Formulas

These closed forms turn a long sum into a single expression. Memorize the first four; understand where the geometric series comes from.

Closed forms

$$\sum_{i=1}^n 1 = n, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

$$\text{Geometric: } \sum_{i=0}^{n-1} r^i = \frac{1-r^{n+1}}{1-r} \quad (r \neq 1), \quad \sum_{i=0}^{\infty} r^i = \frac{1}{1-r} \quad (|r| < 1).$$

$$\text{Arithmetic: } \sum_{i=0}^{n-1} (a + id) = na + d \frac{n(n-1)}{2} \quad (\text{first term } a, \text{ common difference } d).$$

Example 4.1 (The Gauss pairing trick). Why is $\sum_{i=1}^n i = \frac{n(n+1)}{2}$? Write the sum forwards and backwards and add columns:

$$\begin{array}{rcccccc} S & = & 1 & + & 2 & + \cdots + & n \\ S & = & n & + & (n-1) & + \cdots + & 1 \\ \hline 2S & = & (n+1) & + & (n+1) & + \cdots + & (n+1) \end{array}$$

There are n copies of $(n+1)$, so $2S = n(n+1)$ and $S = \frac{n(n+1)}{2}$. (Legend says Gauss found this as a schoolboy.)

Example 4.2 (Using the formulas).

$$\sum_{i=1}^{10} i = \frac{10 \cdot 11}{2} = 55, \quad \sum_{i=1}^5 i^2 = \frac{5 \cdot 6 \cdot 11}{6} = 55, \quad \sum_{i=0}^4 \left(\frac{1}{2}\right)^i = \frac{1 - (1/2)^5}{1 - 1/2} = \frac{31}{16}.$$

Example 4.3 (Infinite geometric series). $\sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$. The series converges only because $|r| = \frac{1}{3} < 1$; for $|r| \geq 1$ it diverges.

Looking ahead. *Summation formulas let us count operations in closed form: a double loop over an $n \times n$ matrix costs $\sum_{i=1}^n n = n^2$ steps — this is how we get the “ $O(n^2)$ ” and “ $O(n^3)$ ” costs in Unit 6. The geometric series also governs how fast iterative methods converge (Units 6, 11, 12).*

5. Exponents and Logarithms

5.1 Exponent laws

For a base $a > 0$ and real exponents m, n :

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n, \quad a^0 = 1, \quad a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}.$$

Example 5.1. $2^3 \cdot 2^4 = 2^7 = 128$, $\frac{x^5}{x^2} = x^3$, $(9)^{1/2} = 3$, $8^{-2/3} = \frac{1}{8^{2/3}} = \frac{1}{(\sqrt[3]{8})^2} = \frac{1}{4}$.

5.2 Logarithms: the inverse of exponentiation

The logarithm answers “to what power must I raise the base?”

$$\log_a x = y \iff a^y = x \quad (a > 0, a \neq 1, x > 0).$$

Two special bases: the **natural log** $\ln x = \log_e x$ (base $e \approx 2.718$) and, in computer science, $\log_2$.

Logarithm laws (each mirrors an exponent law)

$$\log_a(xy) = \log_a x + \log_a y, \quad \log_a \frac{x}{y} = \log_a x - \log_a y, \quad \log_a(x^r) = r \log_a x,$$

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \text{Change of base: } \log_a x = \frac{\ln x}{\ln a}, \quad \text{Inverses: } a^{\log_a x} = x \text{ and } \log_a(a^x) = x.$$

Example 5.2. $\log_2 8 = 3$ since $2^3 = 8$. $\ln(e^5) = 5$. $\log_{10}(1000) = 3$. Using the laws, $\log_2(3 \cdot 8) = \log_2 3 + \log_2 8 = \log_2 3 + 3$.

Example 5.3 (Solving with logs). Solve $5 \cdot 2^x = 160$. Divide: $2^x = 32$, so $x = \log_2 32 = 5$. Solve $e^{3x} = 10$. Take $\ln$: $3x = \ln 10$, so $x = \frac{1}{3} \ln 10 \approx 0.767$.

5.3 What you should recall from MATH 141

You will re-derive these in Unit 5, but keep them handy:

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x} \quad (x > 0), \quad \frac{d}{dx} a^x = a^x \ln a.$$

The number e is special precisely because e^x is its own derivative.

Looking ahead. *Logs turn products into sums — which is why they tame the products of many small probabilities in machine learning (the “log-likelihood” and “cross-entropy loss” of Unit 12). Exponentials define the matrix exponential e^A (Units 4, 11). And $\log_2$ measures how many times you can halve a problem — the source of “ $O(\log n)$ ” running times (Unit 6).*

6. Polynomials

A **polynomial** of degree n is a function

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0,$$

with the a_i constants (*coefficients*) and a_n the *leading* coefficient. A **root** (or *zero*) is a value r with $p(r) = 0$.

Facts about roots

Factor theorem: r is a root $\iff (x - r)$ divides $p(x)$.

Quadratic formula: the roots of $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; the *discriminant* $b^2 - 4ac$ is > 0 (two real roots), $= 0$ (one repeated root), or < 0 (two complex roots).

Counting roots: a degree- n polynomial has *exactly* n roots in $\mathbb{C}$, counted with multiplicity (Fundamental Theorem of Algebra).

Example 6.1 (Factoring). $x^2 - 5x + 6 = (x - 2)(x - 3)$, so the roots are $x = 2$ and $x = 3$. Check with the factor theorem: $p(2) = 4 - 10 + 6 = 0$. ✓

Example 6.2 (Quadratic formula & complex roots). Solve $x^2 + x + 1 = 0$. The discriminant is $1 - 4 = -3 < 0$, so $x = \frac{-1 \pm \sqrt{-3}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ — two complex roots. (This is why we need $\mathbb{C}$; see Unit 3.)

6.1 The binomial theorem

Expanding $(x + y)^n$ uses **binomial coefficients** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Example 6.3. $(x + y)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3 = x^3 + 3x^2y + 3xy^2 + y^3$.

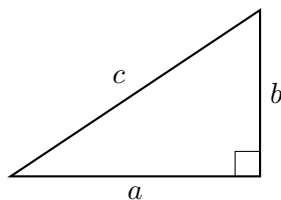
Looking ahead. The characteristic polynomial $\det(A - \lambda I)$ is a degree- n polynomial whose roots are the *eigenvalues* of a matrix (Unit 11). The Fundamental Theorem of Algebra guarantees n of them (possibly complex, possibly repeated) — which is exactly why eigenvalues can be complex even for a real matrix.

7. Geometry: Distance and the Circle

7.1 The Pythagorean theorem

In a right triangle with legs a, b and hypotenuse c ,

$$a^2 + b^2 = c^2.$$



Example 7.1. Legs 3 and 4: $c = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

7.2 The distance formula

Applying Pythagoras to the horizontal and vertical gaps between two points gives the distance in the plane, and the same idea extends to space:

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

$$d((x_1, y_1, z_1), (x_2, y_2, z_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example 7.2. The distance from $(1, 2)$ to $(4, 6)$ is $\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5$.

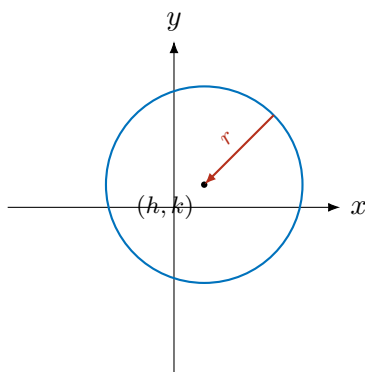
Looking ahead. *This is the formula the course generalizes most. The Euclidean norm $\|x\|_2 = \sqrt{\sum_i x_i^2}$ (Unit 2) is just the distance formula in n dimensions, and least squares (Unit 10) minimizes exactly this distance between a model and the data.*

7.3 The equation of a circle

A circle is the set of points at a fixed distance r (the *radius*) from a center (h, k) . Setting “distance to center = r ” and squaring:

$$(x - h)^2 + (y - k)^2 = r^2.$$

Centered at the origin this is $x^2 + y^2 = r^2$. The 3-D analog is the **sphere** $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$.



Example 7.3. $(x - 1)^2 + (y + 2)^2 = 9$ is the circle with center $(1, -2)$ and radius 3. Conversely, complete the square on $x^2 + y^2 - 6x + 4y + 9 = 0$:

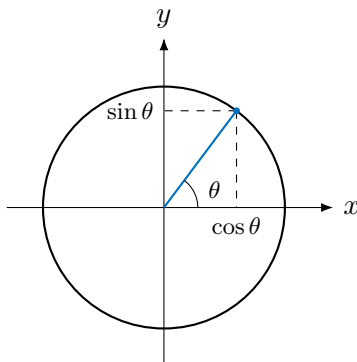
$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = -9 + 9 + 4 \Rightarrow (x - 3)^2 + (y + 2)^2 = 4,$$

a circle of center $(3, -2)$, radius 2.

Looking ahead. *“All points at distance $\leq r$ ” is a disk — and in n dimensions, the set $\{x : \|x\| \leq r\}$ is the unit ball of a norm (Unit 2). Its shape (round for ℓ_2 , diamond for ℓ_1 , square for ℓ_∞) explains why different norms behave so differently in regularization and optimization (Unit 12).*

7.4 A little trigonometry and the unit circle

On the **unit circle** $x^2 + y^2 = 1$, the point at angle θ (measured counterclockwise from the positive x -axis) is $(\cos \theta, \sin \theta)$.



Because the point lies on the circle, we get the **Pythagorean identity**

$$\cos^2 \theta + \sin^2 \theta = 1,$$

and $\tan \theta = \frac{\sin \theta}{\cos \theta}$. Angles are measured in **radians**, where $180^\circ = \pi$; a few values worth knowing:

$$\cos 0 = 1, \sin 0 = 0; \quad \cos \frac{\pi}{2} = 0, \sin \frac{\pi}{2} = 1; \quad \cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

Looking ahead. The angle between two vectors (Unit 2) is measured with $\cos \theta$; cosine similarity in machine learning is exactly this $\cos \theta$. Rotations in the plane, complex multiplication (Unit 3), and quaternions all run on sin and cos.

7.5 Lines in the plane

A non-vertical line has **slope** $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$ and can be written

$$y = mx + b \quad (\text{slope-intercept}), \quad y - y_1 = m(x - x_1) \quad (\text{point-slope}).$$

Example 7.4. The line through $(1, 2)$ and $(3, 8)$ has slope $m = \frac{8-2}{3-1} = 3$, so $y - 2 = 3(x - 1)$, i.e. $y = 3x - 1$.

Looking ahead. A line is the graph of a linear equation in two variables. A system of such equations is the starting point of the entire course (Unit 7), and fitting the “best” line through scattered data is linear regression (Unit 10).

8. Greek Alphabet and Symbol Glossary

Linear algebra borrows heavily from Greek. You do not need to memorize this — just know where to look.

Symbol	Name	Symbol	Name
α	alpha	λ	lambda (<i>eigenvalues</i>)
β	beta	μ	mu (mean)
γ	gamma	σ, Σ	sigma (<i>singular values / sum</i>)
δ, Δ	delta (small change)	π, Π	pi (<i>product</i>)
ε	epsilon (tiny quantity)	ϕ, φ	phi
θ	theta (angle)	ψ	psi
κ	kappa (<i>condition number</i>)	ω, Ω	omega

Symbol	Meaning	Symbol	Meaning
$\in, \notin$	is / is not an element of	$\forall$	for all
$\subseteq, \subsetneq$	subset / proper subset	$\exists$	there exists
$\cup, \cap$	union / intersection	$\Rightarrow$	implies
$\emptyset$	empty set	$\Longleftrightarrow$	if and only if
$\mathbb{R}, \mathbb{C}, \mathbb{Z}$	reals, complex, integers	$\approx$	approximately equals
$ x , \ x\ $	absolute value / norm	$\therefore$	therefore

9. Self-Check (answers below)

Work these with a pencil; if any feel shaky, reread the matching section.

- Write $\{x \in \mathbb{Z} : -2 \leq x < 3\}$ in roster form.
- With $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6\}$, find $A \cup B$, $A \cap B$, $A \setminus B$.
- Solve $|2x - 1| \leq 5$ and write the answer as an interval.
- Evaluate $\sum_{i=1}^6 i$ and $\sum_{i=1}^4 i^2$.
- Evaluate $\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k$.
- Simplify $\log_2 40 - \log_2 5$.
- Solve $3^x = 81$ and $e^{2x} = 7$ (leave the second exact).
- Find the roots of $x^2 - 4x + 13 = 0$.
- Find the distance from $(-1, 3)$ to $(2, -1)$.
- Give the center and radius of $x^2 + y^2 + 4x - 6y + 9 = 0$.

Answers

- $\{-2, -1, 0, 1, 2\}$.
- $A \cup B = \{1, 2, 3, 4, 6\}$, $A \cap B = \{2, 4\}$, $A \setminus B = \{1, 3\}$.

3. $-5 \leq 2x - 1 \leq 5 \Rightarrow -2 \leq x \leq 3$, i.e. $[-2, 3]$.
4. $\sum_{i=1}^6 i = \frac{6 \cdot 7}{2} = 21$; $\sum_{i=1}^4 i^2 = \frac{4 \cdot 5 \cdot 9}{6} = 30$.
5. $\frac{1}{1-1/4} = \frac{4}{3}$.
6. $\log_2 \frac{40}{5} = \log_2 8 = 3$.
7. $x = 4$ (since $3^4 = 81$); $x = \frac{1}{2} \ln 7$.
8. Discriminant $16 - 52 = -36$, so $x = \frac{4 \pm 6i}{2} = 2 \pm 3i$.
9. $\sqrt{(2 - (-1))^2 + (-1 - 3)^2} = \sqrt{9 + 16} = 5$.
10. Complete the square: $(x + 2)^2 + (y - 3)^2 = 4$; center $(-2, 3)$, radius 2.

That is the whole toolkit. If the self-check went well, you are ready for Unit 1. Keep this sheet nearby — every symbol here reappears in the first three weeks.